CISC 473: Deep Learning — Midterm Report

P08: Abstractive Summarization via Parameter-Efficient Fine-Tuning

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Abstract

We investigate whether LoRA placement in BART-base's encoder self-attention, decoder self-attention, or encoder–decoder cross-attention affects summarization quality under exactly matched trainable-parameter budgets. With rank $r = 8$ and query/value projections, each isolated region introduces 147,456 trainable parameters. The implementation now supports controlled CNN/DailyMail training, validation, and held-out CNN/DailyMail and XSum evaluation, with saved configurations and official ROUGE metrics. No genuine training logs, checkpoints, or per-seed results were present in the audited repository. Consequently, the earlier draft's scores and qualitative examples are unverified and are not presented as findings. The main outstanding work is to execute the baseline and placement matrix with three independent seeds, evaluate transfer without further tuning, and conduct the human factual-consistency study.

Introduction and Problem Statement

Abstractive summarization compresses lengthy source documents into fluent, salient, and factually accurate summaries. Pre-trained sequence-to-sequence Transformers such as BART [1] achieve high ROUGE scores by leveraging bidirectional encoder representations coupled with autoregressive decoder generation. However, conventional adaptation requires updating all 139.4 million parameters in BART-base, incurring steep computational overhead and high storage requirements when maintaining separate checkpoints for multiple domains or tasks. Parameter-efficient fine-tuning (PEFT) methods, particularly Low-Rank Adaptation (LoRA) [2], circumvent this cost by freezing the pre-trained weights $W_0 \in \mathbb{R}^{d\times k}$ and injecting trainable rank-decomposition matrices $\Delta W = BA$.

Scope and Rubric Alignment. The primary extension is the matched-budget placement experiment. The supplied P08 guide also requests full fine-tuning, Prefix Tuning, rank and projection ablations, cross-domain evaluation, and a human study. The previous draft described faculty approval for reducing this scope, but supporting feedback was not supplied; that approval remains unverified. The implementation exposes these baseline and ablation options. Their availability in code does not establish that the experiments have been executed.

Research Question. Under strictly matched trainable parameter budgets and identical training conditions, how does the architectural placement of LoRA adapters (encoder self-attention vs. decoder self-attention vs. encoder-decoder cross-attention) govern summarization performance, factual consistency, and zero-shot cross-domain generalization?

Related Work (Updated)

Parameter-efficient fine-tuning for conditional text generation has evolved rapidly, transitioning from uniform parameter perturbation to structural and capacity-aware allocation.

Parameter-Efficient Sequence-to-Sequence Modeling. Large-scale pre-training of sequence-to-sequence architectures, exemplified by BART [1], established denoising objectives as the foundation for modern summarization. While full fine-tuning updates all parameters, Prefix-Tuning [3] introduced continuous virtual prefix vectors prepended to attention keys and values, preserving frozen model weights while achieving competitive generation on XSum with a small task-specific parameter budget. Concurrently, LoRA [2] demonstrated that task-specific parameter updates exhibit low intrinsic dimensionality, parametrizing weight updates via low-rank matrices $\Delta W = BA$. In their foundational work, Hu et al. observed that adapting attention query and value projections suffices for autoregressive language models. Its benchmarks include both language understanding and generation; our experiment specifically tests architectural regions within BART.

Non-Uniform Capacity Allocation and Layer Sensitivity. Recognizing that uniform low-rank parameterization across all layers may be sub-optimal, recent research has explored dynamic allocation. AdaLoRA [4] parameterizes weight matrices using singular value decomposition and iteratively prunes singular values based on importance scores, allocating higher rank to influential matrices. Similarly, WeightLoRA [7] evaluates adapter significance during fine-tuning, demonstrating that adapters are not uniformly useful throughout BART. In a broader theoretical analysis, He et al. [5] established a unified framework connecting adapters, prefix tuning, and LoRA, comparing the design choices shared by several PEFT methods. Furthermore, Valipour et al. [6] introduced DyLoRA to train models across a spectrum of ranks dynamically, supporting selection among multiple ranks without separately training every rank. While these methods demonstrate parameter non-uniformity, they optimize allocation dynamically via complex heuristics, leaving open the empirical question of how distinct, interpretable architectural blocks perform under an identical, static budget.

Factual Consistency in Abstractive Summarization. A known limitation of n-gram overlap metrics such as ROUGE-1, ROUGE-2, and ROUGE-L is their insensitivity to factual hallucinations. Kryscinski et al. [8] demonstrated that high-ROUGE summaries frequently introduce factual inconsistencies unsupported by the source text, proposing FactCC as a weakly supervised evaluation metric. Automatic factuality assessment also has domain limitations. We plan to complement ROUGE with a human study; neither a FactCC experiment nor human judgments are available in this repository.

Method	Year	Key Idea	Relation to Our Work
BART [1]	2020	Bidirectional encoder + autoregressive decoder seq2seq pre-training.	Primary backbone architecture for our summarization study.
LoRA [2]	2022	Low-rank weight update factorization ($\Delta W = BA$).	Core PEFT mechanism evaluated across architectural placements.
Prefix-Tuning [3]	2021	Virtual task-specific prefix vectors prepended to attention keys/values.	Baseline conditional conditioning approach for comparison.
AdaLoRA [4]	2023	SVD-based parameterization with dynamic rank allocation.	Demonstrates non-uniform capacity importance; motivates explicit placement study.
Unified PEFT [5]	2022	Unified functional framework linking adapters, prefixes, and LoRA.	Theoretical grounding for structural placement effects.
DyLoRA [6]	2023	Dynamic search of LoRA ranks during training.	Highlights layer-wise capacity sensitivity in generation tasks.
WeightLoRA [7]	2026	Selectable LoRA units pruned during training.	Confirms adapter utility varies across BART layers.
FactCC [8]	2020	Factual-consistency classifier for abstractive summarization.	Highlights ROUGE limitations and justifies our human evaluation protocol.

Table 1: Comparative taxonomy of related work and foundational methods.

Methodology

Our methodology enforces strict experimental consistency to reduce confounding when assessing whether performance differences originate from adapter placement rather than training variations or capacity disparities.

Model Architecture and Placement Isolation

We use pre-trained `facebook/bart-base` as our backbone, which comprises an encoder with 6 Transformer layers and a decoder with 6 Transformer layers ($d_{\text{model}} = 768$, 12 attention heads, feed-forward dimension $d_{\text{ff}} = 3072$, totaling 139,420,416 parameters). For any weight matrix $W_0 \in \mathbb{R}^{d\times k}$, LoRA parameterizes the forward pass as:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} BAx \tag{1}$$

where $A \in \mathbb{R}^{r\times k}$ uses PEFT's default Kaiming-uniform initialization, $B \in \mathbb{R}^{d\times r}$ is initialized to zero, r is the adaptation rank, and α is a scaling hyperparameter. We set $r = 8$ and $\alpha = 16$, targeting the query (W_q) and value (W_v) projections ($k = d = 768$). Each adapted projection introduces $2 \times 768 \times 8 = 12,288$ parameters.

We construct four distinct LoRA configurations:

1. **Standard LoRA (All Attention):** Injects adapters into W_q and W_v across all 6 encoder self-attention, 6 decoder self-attention, and 6 decoder cross-attention blocks (18 attention layers total $\times 24,576 = 442,368$ trainable parameters, 0.32% of the base model).

2. **Encoder Self-Attention LoRA:** Adapts only `model.encoder.layers.*.self_attn` ($6 \times 24,576 = 147,456$ trainable parameters, 0.11%).
3. **Decoder Self-Attention LoRA:** Adapts only `model.decoder.layers.*.self_attn` ($6 \times 24,576 = 147,456$ trainable parameters, 0.11%).
4. **Cross-Attention LoRA:** Adapts only `model.decoder.layers.*.encoder_attn` ($6 \times 24,576 = 147,456$ trainable parameters, 0.11%).

Crucially, configurations (2), (3), and (4) share an identical parameter budget of exactly 147,456 parameters, controlling adapter capacity in the architectural comparison.

Training Setup

To ensure fair and reproducible comparisons, the three isolated placement variants are configured with matched hyperparameters as detailed in Table 2. We use the AdamW optimizer with a linear learning rate warmup followed by linear decay. Effective batch size is maintained at 32 across all runs via gradient accumulation.

Hyperparameter	**Value**
Backbone checkpoint	`facebook/bart-base` (139.4M parameters)
Optimiser	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 1 \times 10^{-8}$)
Learning rate	3×10^{-4} (LoRA variants); 3×10^{-5} (Full Fine-Tuning)
Learning rate schedule	Linear warmup (500 steps) with linear decay
Batch size	8 per device $\times$ 4 gradient accumulation steps = 32 effective
Training epochs	3 epochs
Weight decay	0.01
Precision	FP32 by default; optional CUDA FP16; gradient clipping ($\max \|\mathbf{g}\|_2 = 1.0$)
Generation beam search	Beam width = 4, length penalty = 2.0, min length = 56, max length = 128
Repetition penalty	3-gram blocking (`no_repeat_ngram_size = 3`)
Planned independent training seeds	42, 123, 999

Table 2: Standardized training and generation hyperparameters.

Dataset and Preprocessing

We prepare in-domain summarization on the CNN/DailyMail dataset (Version 3.0.0), comprising 287,113 training, 13,368 validation, and 11,490 test document-summary pairs. Source news stories are truncated to a maximum of 1,024 BPE tokens, and target multi-sentence summaries are capped at 128 tokens. Special tokens (`<s>`, `</s>`) are prepended and appended, with target padding tokens replaced by -100 to exclude them from cross-entropy loss computation. For zero-shot cross-domain evaluation, we use the test split of XSum (226,711 total samples: 204,045 train, 11,332 validation, 11,334 test), which features highly abstractive, single-sentence summaries of BBC news articles.

Preliminary Results

Implementation Evidence and Outstanding Measurements

Only architecture-derived parameter budgets can currently be reported. Each projection adds $r(d_{\text{in}}+d_{\text{out}})$ parameters. For six attention blocks, two projections per block, $r = 8$, and $d = 768$, this is 147,456 for each isolated placement; adapting all three regions adds 442,368. These correspond to 0.1058% and 0.3173% of the 139,420,416-parameter base model, respectively. Percentages in saved run manifests use total parameters *including* adapters as the denominator.

Configuration	Expected trainable parameters	ROUGE / training evidence
Zero-shot BART-base	0	Unavailable
Full fine-tuning	139,420,416	Unavailable
All-attention LoRA	442,368	Unavailable
Encoder self-attention LoRA	147,456	Unavailable
Decoder self-attention LoRA	147,456	Unavailable
Cross-attention LoRA	147,456	Unavailable
Prefix Tuning (20 tokens)	184,320	Unavailable

Table 3: Architecture budgets are distinct from unexecuted quality experiments.

Evaluation Protocol

The default pilot uses a fixed, shuffled 10,000-example CNN/DailyMail training subset (data seed 42) and the full official validation split. The data seed stays fixed while training seeds vary over 42, 123, and 999. The final checkpoint is selected by epoch/update count, not by test performance. Test runs load that checkpoint and evaluate the full official CNN/DailyMail and XSum test splits without training. Hub revision SHAs, split sizes, selected-example hashes, generation settings, package versions, and model configuration are saved.

ROUGE-1, ROUGE-2, and ROUGE-L use Hugging Face Evaluate's official ROUGE implementation with stemming and arithmetic mean per-example F1, multiplied by 100. ROUGE-L is distinct from ROUGE-Lsum. References for ROUGE retain their full original text; training and loss labels are truncated to 128 tokens. Dynamic label padding uses -100. The generation settings in Table 2 apply to both test domains. Keeping the CNN/DailyMail minimum generation length of 56 on single-sentence XSum is a protocol choice that may disadvantage transfer; any alternative should be declared as a separate evaluation ablation.

The aggregation script requires one completed run for each expected seed and rejects duplicates, missing seeds, or incompatible within-configuration protocols. It computes mean and sample standard deviation ($n - 1$ denominator) from saved results. Full fine-tuning, Prefix Tuning, and all-attention LoRA are comparison baselines, not matched-parameter placement variants. Equal examples and updates do not imply equal FLOPs or wall-clock compute; timing and update counts must be considered when assessing the guide's matched-compute objective.

Unverified Claims Removed from the Earlier Draft

There is no saved evidence supporting the earlier zero-shot, full fine-tuning, all-attention, or isolated-placement ROUGE means or standard deviations. The 42.30/19.50/39.40 triplet attributed to a published BART-base baseline lacks a verified matching source and protocol. The baseline-reproduction claim, 98.9% retention claim, relative placement improvements, and conclusions about functional specialization therefore remain unverified. The FAA qualitative example has no dataset

ID or checkpoint provenance and was removed. The hard-coded score plot was also removed. No ranking of placements can yet be concluded.

Challenges and Adaptations

The audit found that training did not connect ROUGE to the Trainer, an evaluation fallback invented an approximation for ROUGE-L, and the plotting script contained hard-coded scores. Generation settings were only partially applied, metrics and final models were not explicitly saved for downstream evaluation, and test-set and XSum paths were absent. These implementation defects are addressed by shared model construction, official metrics with failures propagated, explicit generation configuration, persistent run manifests, and evaluation-only checkpoint loading.

The previous draft's claims of observed CUDA out-of-memory failures, successful GPU tuning, and earlier unit-test verification were not accompanied by logs or tests. They are not retained as observations. Actual checks and their outcomes are recorded in the repository's audit documentation. Full training requires separate execution; smoke-test outputs do not constitute research results.

Baseline reproduction and preliminary ablations are necessary for the midterm rubric's quantitative-results criteria. A functional implementation alone does not satisfy those criteria. The exact faculty-approved scope, any external logs, and author contribution declarations must be confirmed by the project team.

Updated Week 7–10 Plan

Table 4 outlines our roadmap for the remaining project period.

Week	Tasks	Responsible
7	Evaluate trained models on XSum test set for zero-shot cross-domain generalization; conduct training throughput (tokens/sec) and peak memory profiling.	Wen Bin / Junn Rong
8	Conduct human factual consistency evaluation (10 participants, 20 summaries per configuration); calculate inter-annotator agreement (Fleiss' κ); run FactCC metric.	Junn Rong / Wen Bin
9	Synthesize ROUGE, factuality, and efficiency results; generate parameter-performance Pareto trade-off curves; conduct qualitative failure case analysis.	Both
10	Final report drafting, comprehensive code clean-up, reproduction verification from a clean environment, and presentation deck preparation. **(Final due)**	Both

Table 4: Updated project schedule and task allocation for Weeks 7–10.

Preliminary Novel Contribution (Optional Bonus)

The proposed contribution is a controlled study of three interpretable LoRA placements with exactly matched adapter capacity. Architectural symmetry makes this comparison feasible, but it does not establish which placement performs best. Any claim of improved quality, efficiency, transfer, or factuality must await completed multi-seed experiments and the planned human study. A claim of novelty also requires continued comparison with the literature.

Contribution Declaration

The original team declaration assigned 50% to Choo Wen Bin and 50% to Nai Junn Rong. These author-provided percentages are retained as declarations, not as verified activity logs. The team should update actual contributions before submission.

AI Usage Disclosure

An OpenAI Codex assistant was used for this code audit, implementation corrections, test and documentation preparation, and revision of unsupported report claims. AI-generated code and prose require author review. No model-training measurements, human judgments, or experimental observations were generated to fill missing results. This disclosure covers the current revision; the team must disclose any earlier AI use separately and confirm the final wording before submission.

References

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